

Chi-square is a statistical tool that helps us to decide if the observed ratio is close enough to the expected ratio to be acceptable. Chi-square analysis can be used in any area, not just genetics. Whenever you have to determine if an expected ratio fits an observed ratio, you can use the Chi-square.

$$X^2 = \sum \frac{(O-E)^2}{E}$$

Chi Square Significance Table

Degrees of Freedom (n)	5% Probability Value (P)
1	3.84
2	5.99
3	7.81
4	9.49

1. In peas, yellow seeds (A) are dominant over green (a) seeds. In a cross between two plants both heterozygous for seed color, the following was observed:

Yellow = 4400

Green = 1624

Does the data fit the predicted phenotypic ratio?

Phenotype	Observed (O)	Expected (E)	O-E	(O-E) ²	$\frac{(O-E)^2}{E}$
Totals					

2. In peas, smooth seeds (R) are dominant over wrinkled (r) seeds. In the P generation, a plant homozygous for smooth seeds is crossed with a plant with wrinkled seeds. The resulting F₁ plants are crossed. The seeds of the observed F₂ generation were:

Smooth = 5474

Wrinkled = 1850

Does the data fit the predicted phenotypic ratio?

Phenotype	Observed (O)	Expected (E)	O-E	(O-E) ²	$\frac{(O-E)^2}{E}$
Totals					

3. In a flowering plant, white flowers (B) are dominant over red (b), and short plants (E) are dominant over tall (e) plants. When two double heterozygote (BbEe) plants were crossed, the resulting phenotypes were observed:

White, short = 206

Red, short = 83

White, tall = 65

Red, tall = 30

Does the data fit the predicted phenotypic ratio?

Phenotype	Observed (O)	Expected (E)	O-E	(O-E) ²	$\frac{(O-E)^2}{E}$
Totals					

4. In corn, purple kernels (D) are dominant over yellow (d), and smooth kernels (G) are dominant over shrunk (g). An ear of corn has 381 kernels, illustrated at right:

A: purple, smooth = 216

B: purple, shrunk = 79

C: yellow, smooth = 65

D: yellow, shrunk = 21

Does the data fit your predicted phenotypic ratio?

(Your prediction would be the kernels are the result of a double heterozygous cross with a ratio of 9:3:3:1)



Phenotype	Observed (O)	Expected (E)	O-E	(O-E) ²	$\frac{(O-E)^2}{E}$
Totals					

6. Color blindness is a sex-linked trait in Bombats. A female who is a carrier of the color blind allele mates with a male who is color blind. The phenotypes of their offspring are:

Normal female = 132

Color blind female = 124

Normal male = 126

Color blind male = 136

Does the data fit your predicted phenotypic ratio?

Phenotype	Observed (O)	Expected (E)	O-E	(O-E) ²	$\frac{(O-E)^2}{E}$
Totals					

7. In cats, fur color is determined by the codominant, sex-linked alleles: black (B) and orange (O). A calico female ($X^B X^O$) is bred (many times) with a black male ($X^B Y$). They produce the following offspring:

Black female = 78

Calico female = 65

Black male = 81

Orange male = 45

Does the data fit your predicted phenotypic ratio?

Phenotype	Observed (O)	Expected (E)	O-E	(O-E) ²	$\frac{(O-E)^2}{E}$
Totals					